

7(a)

- Emission from radioactive daughter products
- Self absorption in source
- Absorption in air before reaching detector

2018 JC2 H2 Physics Prelims Exam Solution (9749 Paper 3)

- Detector not sensitive to all radiations
- Window/detector may absorb some radiations
- Dead-time of the counter
- Background radiation

Any two

7 (b)(i)

Curve is not smooth

Or

Curve fluctuates/is jagged

7 (b)(i)

Evidence of background radiation taken account [B1]

Half life determine twice [B1]

Half life = 1.5h (accepting a range of 1.4-1.6 [A1])

7 (c)

Half life has no change as radioactive is spontaneous/is not affected by external factor

8 (a)